

```

function ADD WEIGHT( $w$ )
  for  $s \in \{s \mid \tau_s \in \phi\}$  do
    if  $|\tau_{s_i}| > |\vec{A}_{s_i}|$  then  $\triangleright$  # of activations  $>$  than # of distributions
      INCREASE( $\alpha_{s_i}, |\tau_{s_i}|$ )
      INCREASE( $\beta_{s_i}, |\tau_{s_i}|$ )
    end if
     $count \leftarrow 0$ 
    while  $count < |\tau_{s_i}|$  do
       $\alpha, \beta \leftarrow \text{ORDERED}(\alpha_{s_i}, \beta_{s_i})[count]$ 
       $ratio_\alpha \leftarrow \alpha / (\alpha + \beta)$ 
       $ratio_\beta \leftarrow \beta / (\alpha + \beta)$ 
       $\alpha \leftarrow \alpha + w * ratio_\alpha$ 
       $\beta \leftarrow \beta + w * ratio_\beta$ 
       $count \leftarrow count + 1$ 
    end while
  end for
end function

```